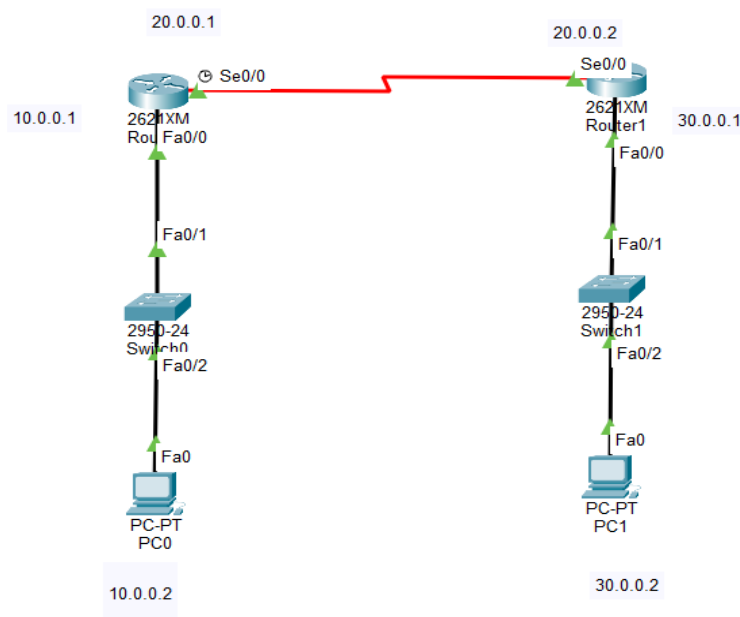


2 : Introduction to VPNs Theory :Types of VPNs: Site-to-site, remote access.

Practical :Configure a basic site-to-site VPN in Packet Tracer. Task: Submit configuration steps for a VPN.

Task 1.

In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.



```
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=3ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255
Reply from 30.0.0.1: bytes=32 time<1ms TTL=255

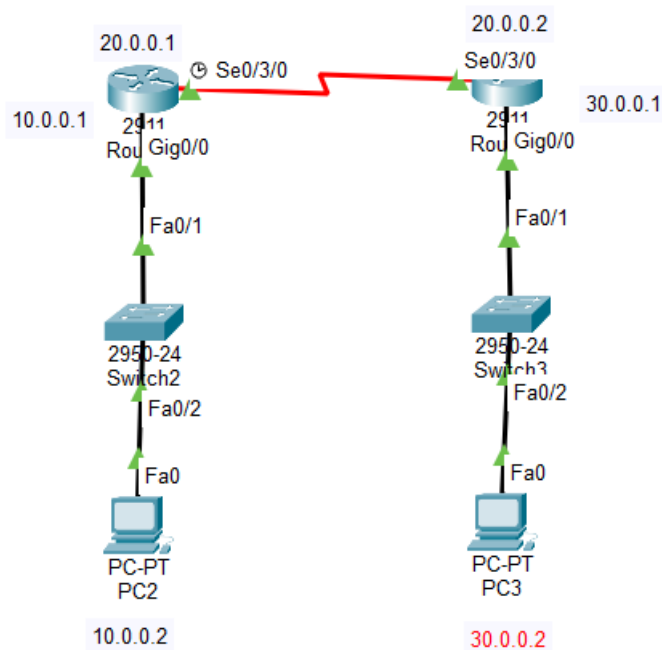
Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 2.

Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.



```
C:\>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time=2ms TTL=254
Reply from 30.0.0.1: bytes=32 time=1ms TTL=254
Reply from 30.0.0.1: bytes=32 time=1ms TTL=254
Reply from 30.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

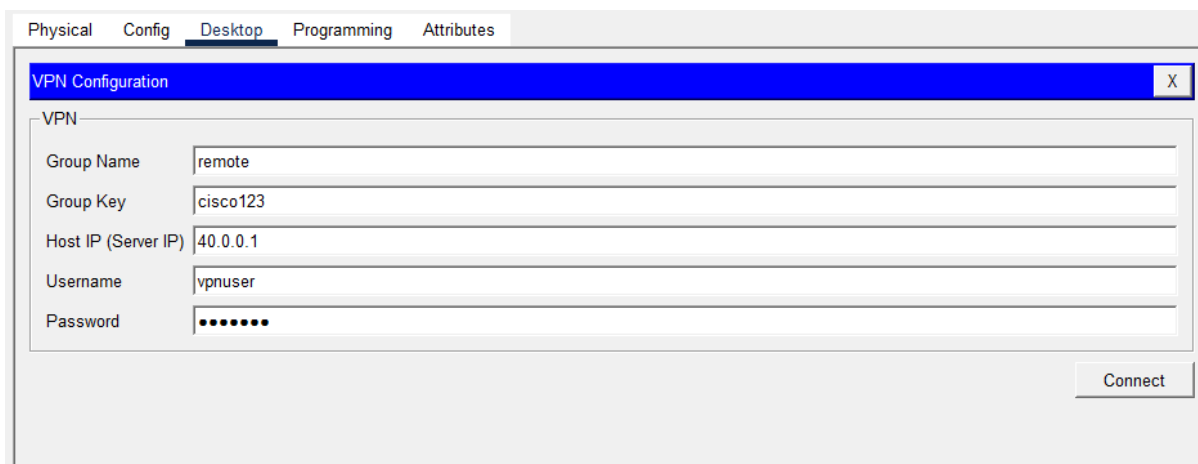
Request timed out.
Reply from 30.0.0.2: bytes=32 time=4ms TTL=126
Reply from 30.0.0.2: bytes=32 time=5ms TTL=126
Reply from 30.0.0.2: bytes=32 time=3ms TTL=126

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms
```

Task 3.

Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.

Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.



The screenshot shows the 'VPN Configuration' window in Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is selected. The VPN Configuration window has a title bar with 'VPN Configuration' and a close button. Below the title bar is a 'VPN' section with the following fields:

Field	Value
Group Name	remote
Group Key	cisco123
Host IP (Server IP)	40.0.0.1
Username	vpnuser
Password	*****

A 'Connect' button is located at the bottom right of the window.

Field	Value
Host IP	40.0.0.1
Group Name	REMOTE
Group Key	CISCO123
Username	vpnuser
Password	vpnpass

Task 4.

List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

Point	Site-to-Site VPN	Remote Access VPN
Purpose	Connects two complete networks/branches	Connects an individual user/device to a network
Connection	Router/firewall ↔ Router/firewall	PC/laptop ↔ VPN gateway
Users	Many users/devices at both branches	Usually one remote user/device
Configuration	Configured mainly on VPN routers/firewalls	VPN client configured on the user's device
Example	Branch A ↔ Branch B	Employee working from home ↔ company network
Our practical	R0 ↔ R1 using GRE Tunnel0	PC VPN client ↔ branch router
Authentication	Usually device-to-device authentication	Usually username/password and/or certificates
Advantages	1. Connects entire branch networks.2. Users don't need individual VPN clients.	1. Allows users to work remotely.2. Flexible and convenient for mobile users.
Disadvantages	1. Requires VPN-capable routers/firewalls at both sites.2. Configuration can be more complex.	1. Each user/device requires VPN configuration.2. Managing many remote users can become difficult.

Task 5.

Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

1. R0 — LAN interface

enable

configure terminal

```
interface gigabitEthernet 0/0
```

```
ip address 10.0.0.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

This configures the LAN gateway for Branch 1.

2. R0 — WAN interface

```
interface serial 0/3/0
```

```
ip address 20.0.0.1 255.255.255.252
```

```
clock rate 64000
```

```
no shutdown
```

```
exit
```

This provides the WAN connection between R0 and R1.

3. R0 — GRE Tunnel

```
interface tunnel 0
```

```
ip address 172.16.0.1 255.255.255.252
```

```
tunnel source serial 0/3/0
```

```
tunnel destination 20.0.0.2
```

```
tunnel mode gre ip
```

```
no shutdown
```

```
exit
```

This creates the GRE tunnel endpoint on R0.

4. R0 — Route to Branch 2

```
ip route 30.0.0.0 255.255.255.0 172.16.0.2
```

This tells R0 to send traffic for Branch 2 through the GRE tunnel.

R1 Configuration

5. R1 — LAN interface

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0
```

```
ip address 30.0.0.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

This configures the LAN gateway for Branch 2.

6. R1 — WAN interface

```
interface serial 0/3/0
```

```
ip address 20.0.0.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

This connects R1 to R0 through the WAN.

7. R1 — GRE Tunnel

```
interface tunnel 0
```

```
ip address 172.16.0.2 255.255.255.252
```

```
tunnel source serial 0/3/0
```

```
tunnel destination 20.0.0.1
```

```
tunnel mode gre ip
```

```
no shutdown
```

```
exit
```

This creates the second endpoint of the GRE tunnel.

8. R1 — Route to Branch 1

```
ip route 10.0.0.0 255.255.255.0 172.16.0.1
```

This tells R1 to send traffic for Branch 1 through the GRE tunnel.

9. Verification Commands

Check interfaces

On both routers:

```
show ip interface brief
```

Expected:

R0:

```
Serial0/3/0  20.0.0.1  up/up
```

```
Tunnel0     172.16.0.1  up/up
```

R1:

```
Serial0/3/0  20.0.0.2  up/up
```

```
Tunnel0     172.16.0.2  up/up
```

Check the GRE tunnel

From R0:

```
ping 172.16.0.2
```

From R1:

ping 172.16.0.1

Expected:

!!!!

Success rate is 100 percent

Check routes

R0:

show ip route

R1:

show ip route

You should see the static routes to the opposite LAN.

10. Final PC-to-PC test

From **PC0**:

ping 30.0.0.2

From **PC1**:

ping 10.0.0.2

Successful replies confirm that traffic is passing between the two branch networks through the GRE tunnel.